

SIMATS ENGINEERING

ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence | COURSE CODE: CSA17

RESOLUTION ALGORITHM IMPLEMENTATION – ANSWERS

COURSE OUTCOME COVERED: CO3 – Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, and logical reasoning for valid conclusions and decision making.

Assessment Tool Weightage: 20% **Time Duration:** 1 Hour **Total Questions:** 5 **Total Marks:** 30

Code of Conduct: I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

Signature of the student: _____

QUESTION 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal: Prove that the ground is wet using the Resolution Algorithm.

1. Propositional Logic

Let:

R = It is raining

W = Ground is wet

Rule:

$R \rightarrow W$

Fact:

R

2. CNF Conversion

$R \rightarrow W$

$= \neg R \vee W$

Fact:

R

CNF clauses:

C1 = $\neg R \vee W$

C2 = R

3. Negate the Goal

Goal: W

Negated goal: $\neg W$

C3 = $\neg W$

4. Resolution Algorithm

Resolve C1 and C2:

$(\neg R \vee W)$

(R)

■■■■■■■■■■

W

Now resolve W with the negated goal:

(W)

($\neg W$)

■■■■■■■

■

Empty clause ■ is obtained.

5. Conclusion

Since the empty clause ■ is derived, the negated goal is contradictory. Therefore, **the ground is wet** is proved.

QUESTION 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal: Prove that Rahul receives internal marks.

1. Propositional Logic

Let:

S = Rahul submitted the assignment

M = Rahul receives internal marks

Rule:

$S \rightarrow M$

Fact:

S

2. CNF Conversion

$S \rightarrow M$

$= \neg S \vee M$

Fact:

S

CNF:

C1 = $\neg S \vee M$

C2 = S

3. Negate the Goal

Goal: M

Negated goal: $\neg M$

C3 = $\neg M$

4. Resolution Algorithm

Resolve C1 and C2:

$(\neg S \vee M)$

(S)

■■■■■■■■■■

M

Resolve M with $\neg M$:

(M)

$(\neg M)$

■■■■■■■

■

5. Conclusion

The empty clause ■ is obtained. Hence, **Rahul receives internal marks** is proved.

QUESTION 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal: Prove that Priya can borrow books.

1. Propositional Logic

Let:

L = Priya is a library member

B = Priya can borrow books

Rule:

$L \rightarrow B$

Fact:

L

2. CNF Conversion

$L \rightarrow B$

$= \neg L \vee B$

Fact:

L

CNF:

C1 = $\neg L \vee B$

C2 = L

3. Negate the Goal

Goal: B

Negated goal: $\neg B$

C3 = $\neg B$

4. Resolution

Resolve C1 and C2:

$(\neg L \vee B)$

(L)

■■■■■■■■■■

B

Now resolve B with $\neg B$:

(B)

($\neg B$)

■■■■■■

■

5. Final Conclusion

The empty clause ■ proves that the negated goal is false. Therefore, **Priya can borrow books.**

QUESTION 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal: Prove that Arun is eligible for placement.

1. Propositional Logic

Let:

A = Arun cleared the aptitude test

E = Arun is eligible for placement

Rule:

$A \rightarrow E$

Fact:

A

2. CNF Clauses

$A \rightarrow E$

$= \neg A \vee E$

C1 = $\neg A \vee E$

C2 = A

3. Negate the Goal

Goal: E

Negated goal: $\neg E$

C3 = $\neg E$

4. Resolution Steps

Step 1:

$(\neg A \vee E)$

(A)

■■■■■■■■■■

E

Step 2:

(E)

$(\neg E)$

■■■■■■■

■

Empty clause ■ is obtained.

5. Resolution Tree

```

   $\neg A \vee E$ 
 /   \
A      E
 |
 $\neg E$ 
 |
  ■
```

Therefore E is proved.

6. Conclusion

Since ■ is derived, **Arun is eligible for placement.**

QUESTION 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal: Prove that the user is granted access.

1. Propositional Logic

Let:

P = User entered the correct password

A = User is authenticated

G = User is granted access

Rules:

$P \rightarrow A$

$A \rightarrow G$

Fact:

P

2. CNF Conversion

Rule 1:

$P \rightarrow A$

$= \neg P \vee A$

Rule 2:

$A \rightarrow G$

$= \neg A \vee G$

Fact:

P

CNF:

C1 = $\neg P \vee A$

C2 = $\neg A \vee G$

C3 = P

3. Negate the Goal

Goal: G

Negated goal: $\neg G$

C4 = $\neg G$

4. Resolution Steps

Step 1:

$(\neg P \vee A)$

(P)

■■■■■■■■■■

A

Step 2:

$(\neg A \vee G)$

(A)

■■■■■■■■■■

G

Step 3:

(G)

(¬G)

■■■■■■■

■

Empty clause ■ is obtained.

5. Explanation

The correct password first proves that the user is authenticated. Authentication then proves that access is granted. Finally, the derived fact Granted Access contradicts the negated goal ¬Granted Access, producing the empty clause.

6. Final Conclusion

Correct Password

↓

Authenticated

↓

Granted Access

Therefore:

■ is derived.

The user is granted access.

END OF ASSESSMENT TOOL 4 ANSWERS